

SURYA SINGH

Backend Engineer | Python, FastAPI, Docker, AWS | REST API Design & Cloud Deployment

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PROFESSIONAL SUMMARY

Backend engineer with proven experience designing and deploying production systems on AWS EC2. Built real-time event processing services, secure REST APIs with JWT authentication, and scalable database architectures. Strong foundation in Python, FastAPI, and cloud infrastructure.

PROJECTS

GitHub OSS Issue Notifier | 🐙 Github

Developer

- **Developed real-time issue monitoring service** Tracks 10+ repositories and processes 100+ issue events daily. Saves ~2 hours/week of manual tracking.
 - **Deployed on AWS EC2** using **Docker**; automated deployment pipeline via **GitHub Actions**—any code push triggers production update.
 - Followed **SDLC** practices from design through deployment.
 - Integrated GitHub REST API to monitor 10+ repos.
 - Debugged and resolved **CI/CD pipeline** failures — permission errors, runner registration, environment misconfig.
- Technologies / Tools Used :** Python, FastAPI, Docker, AWS EC2, GitHub Actions

CGAN Synthetic Financial Data Generator | 🐙 Github 🌐 Live-Demo

Developer

- **Developed** a **Conditional GAN (CGAN)** to generate synthetic fraud transaction data and address class imbalance in the IEEE-CIS dataset.
 - **Generated 5,000** synthetic transactions and validated data quality through feature distribution and drift analysis.
 - **Achieved 97.25% accuracy, 96% precision, and 0.80 F1-score** using **Random Forest** evaluation; **debugged** model convergence and data leakage issues during training, deployed results via a Streamlit dashboard.
 - **Collaborated** with 4 teammates on model development, participated in code reviews and version-controlled work using Git.
- Technologies / Tools Used :** Python, Pytorch, Pandas, Scikit-learn, Numpy

Personal Portfolio & Deployment Pipeline | 🌐 Website

Developer

- **Built and deployed** a **FastAPI + Jinja2** portfolio site with custom domain, SSL/TLS via Let's Encrypt, and Nginx reverse proxy on **AWS EC2**.
 - **Automated deployment** via GitHub Actions **CI/CD** — every push rebuilds and redeploys the Docker container to production with zero manual steps.
 - **Registered and configured** a custom domain with **DNS routing**, serving **RESTful** endpoints over HTTPS through a production Nginx reverse proxy setup.
 - **Wrote unit tests** using pytest to validate API endpoints and core logic
- Technologies / Tools Used :** FastAPI, Python, Docker, AWS EC2, Nginx, GitHub Actions, SSL/TLS, pytest

SmartRecruitAI (Resume Analyzer) | 🐙 Github

Developer

- **Developed** a resume analysis backend utilizing **Python** and **NLP** techniques to extract candidate information.
 - **Integrated third-party Gemini API** for recommending job roles based on extracted skills and experience.
 - **Automated** the resume screening workflow, enhancing recruitment efficiency.
- Technologies / Tools Used :** Python, NLP, Gemini API

EDUCATIONS

Naraina College of Engineering & Technology | Kanpur

B.Tech CSE

2023 - 2026

CGPA : 6.72

SKILLS

Programming Languages :	Python, SQL
Frameworks & Libraries :	FastAPI
Databases :	PostgreSQL, Supabase, MySQL
Tools & Platforms :	GitHub Actions, Git, GitHub, Linux, Docker, AWS EC2, CI/CD, Nginx, JWT Authentication, SSL/TLS
ML Libraries :	Pandas, NumPy, Scikit-learn, PyTorch
Core CS Fundamentals: :	DBMS, OS, Computer Networks, OOP

ACHIEVEMENTS

- **Active GitHub:** Contributed 400+ commits (2024–2025) through personal projects and open-source contributions.
- **Hackathon Participant:** Engaged in the Adobe India Hackathon 2025 and attended AI/ML and Blockchain workshops at Cognizance IIT-Roorkee.

TECHNICAL WRITING

- [Building an OSS Issue Notifier with FastAPI, Docker & AWS.](#)
- [Generating Synthetic Financial Data Using CGANs.](#)